

SEC P vs NP — notebook summary

SEC (autonomous) — attributed to Ludovico Kubler

2026-05-12 04:30 UTC

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Generated 2026-05-12 04:30 UTC

- Cycles recorded: **639**
- Time span: 438.8h (~1.46 cycles/h)
- Notebook: /home/ludo/Scrivania/SEC/research/pvsnp_notebook.jsonl

Verdict distribution

Verdict	Count
INCONCLUSIVE	602
BARRIER_HIT	18
FALSIFIED	15
SUPPORTED	4

Mathematical fields explored (field_A)

Field	Cycles
Real Algebraic Geometry	34
Schur-Weyl Duality	22
Representation Theory of Symmetric Groups	17
Matroid Theory	15
Free Probability	14
Additive Combinatorics	10
Free Probability Theory	10
Noncommutative L^p Geometry	8
Polymatroid Theory	8
Algebraic Geometry	7
Random Matrix Theory	7
Finite Geometry	6
Spectral Graph Theory	6
Plethysm Theory	6
Ergodic Circuit Framework (communication complexity via dynamical systems)	4
{“framework_name”: “Ergodic Circuit Framework”, “math_branch”: “COMM_COMPLEXITY”}	4
Fourier Analysis on Boolean Functions	4
Fourier Analysis of Boolean Functions	4
Algebraic Geometry of Secant Varieties	4

Field	Cycles
FOURIER_ANALYSIS	4
Noncommutative Geometry	4
Fourier analysis of boolean functions	3
Algebraic Topology	3
matroid theory	3
Coarse Geometric Lifting (CGL) — Geometric Complexity Theory	3
{“framework_name”: “Tropical Circuit Weight Analysis (TCWA)”, “math_branch”: “BOUNDED_ARITHMETIC”}	3
Tropical Circuit Weight Analysis (BOUNDED_ARITHMETIC)	3
Schur-Weyl duality	3
Diophantine Approximation	3
Persistent Homology	3

Last 15 cycles (chronological)

Time	Verdict	Title
2026-05-11 21:47 UTC	INCONCLUSIVE	Schur-Weyl Multiplicity Gap in Permanent vs Determinant Decomposi
2026-05-11 21:57 UTC	INCONCLUSIVE	Dual Convex Body Width Bounds SOS Degree for Max-CUT
2026-05-11 22:27 UTC	INCONCLUSIVE	Cycle Matroid Rank Inverse Proportional to Resolution Proof Size
2026-05-11 22:59 UTC	INCONCLUSIVE	Cohomological Dimension Inverse Proportional to Disjointness Comm
2026-05-11 23:12 UTC	INCONCLUSIVE	Kronecker Coefficient Gap in Set Disjointness
2026-05-11 23:33 UTC	INCONCLUSIVE	Communication Compl Symmetric Tensor Rank Gap in Permanent vs Determinant Decompositi
2026-05-12 01:18 UTC	INCONCLUSIVE	Free Cumulant Sum Bounded by Disjointness
2026-05-12 01:34 UTC	INCONCLUSIVE	Communication Complexit Plethysm Coefficient Gap in Permanent vs Determinant Decompositio
2026-05-12 01:48 UTC	INCONCLUSIVE	Symmetric Group Fourier Coefficients Inverse Proportional to ACC ⁰
2026-05-12 01:54 UTC	INCONCLUSIVE	Integer Point Count in Solution Polytope Bounds Extended Frege Pr
2026-05-12 02:14 UTC	INCONCLUSIVE	Matroid Rank Bounded by Nisan-Wigderson Seed Length for 3-SAT

Time	Verdict	Title
2026-05-12 02:20 UTC	INCONCLUSIVE	Matroid Rank Submodularity and Monotone Circuit Size for k-CLIQUE
2026-05-12 02:46 UTC	INCONCLUSIVE	Symplectic Rank of Disjointness Communication Matrix Bounds Rando
2026-05-12 03:10 UTC	INCONCLUSIVE	Schur Coefficient Density Inverse Proportional to Frege Proof Len
2026-05-12 03:25 UTC	INCONCLUSIVE	Free Cumulant Sum Bounded by Log-Size for Read-Twice BPs

How to read the reports

- reports/supported_findings.md — SUPPORTED conjectures with full test code. Paper-quality.
- reports/falsified_findings.md — FALSIFIED with counterexamples.
- reports/daily_YYYY-MM-DD.md — chronological log by day.
- pvsnp_notebook.jsonl — raw JSONL, one entry per cycle (source of truth).

PDF export

Install pandoc + a LaTeX engine once (`sudo apt install pandoc texlive-xetex`), then:

```
cd research/reports
for f in supported_findings.md falsified_findings.md notebook_summary.md; do
  pandoc "$f" -o "${f%.md}.pdf" --pdf-engine=xelatex -V geometry:margin=2cm
done
```